

A Feasible IoT-Based System for Precision Agriculture

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Abstract—This paper presents a feasible and a low-cost IoT based monitoring system for precision agriculture, with emphasis to viticulture. A field/crop data are acquired by self-powered measuring station and sent to remote collector, located in the home or office. The collector accepts the data and forward them via home/office WiFi to commercial or open IoT servers (as ThingSpeak) for further visualization, analyze and react back. Using available APIs, any web application on any server can communicate with IoT server, and thus with field station. Every authorized person or farmer, who has computer or gadget with Internet access can monitor parameters of interest and control back associated processes as irrigation or ventilation. Principle of operation, hardware and software architecture as well as preliminary results are shortly elaborated.

Keywords—IoT; LoRa; agriculture; telemonitoring; remote control; node MCU; precision

I. INTRODUCTION

Global world growing population faces many challenges, among others, in food production. There are obvious efforts to employ Industry 4.0 technologies in agriculture, in order to achieve increased production in term of quantity and quality, efficient ecological management, financial savings, water supply management and other benefits.

Specific data for precision agriculture (PA) are provided by range of specific sensors connected to hardware devices with adequate software applications [2]. Such systems are capable to measure, visualize, monitor and analyze soil and environmental parameters [3] and provide adequate feedbacks from the user [1,2,5,6]. There are a large number of such systems on the market, with different characteristics and prices. The set of parameters to be measured varies from temperature, humidity to sunlight as well as to content of chlorophyll in the leaf. Sometimes, the chemical elements in the soil are included. Also, their working principle varies from stand-alone components with manually readings to those that have their own network and are powered by custom designed applications on dedicated servers.

However, the designing PA system for purpose of viticulture presents research and engineering challenge [1]. The wine yards are of large areas, often away from home or office and it is not easy to be covered by reliable communication network. WiFi that is the most used networking protocol in PA is suitable for small plots or indoor farming. Usually requires the presence of electricity or more powerful solar sources on the plantation, because the consumption of the router network is not small.

Taking into account the specifics of viticulture and wine yards concerning PA, this paper proposes an adequate system, which can meet the requirements. Being inexpensive, based on available and open-source technologies, scalable and dynamic, such system is a feasible solution for developing countries, where the farmers have not possibility to deal with expensive commercial solutions. It offers several functional advantages as: (i) powering autonomy, (ii) long-range communication, (iii) integration with existing commercial or free IoT platforms like ThingSpeak, (iv) easy embedding in individual or company web sites/servers and (v) accessibility with any person or farmer equipped by internet access. The same system and approach can be employed in similar PA applications related to other crops.

Here we describe some elements of proposed PA solution: operating principle, hardware-software architecture and preliminary testing results.

II. PRINCIPLE OF OPERATION

A. Hardware considerations

The operation principle follows system architecture given in Figure 1. Autonomy SMART-MES STATION integrates: solar panel, power supply electronics with internal battery, ESP32 based Node MCU (from Heltec) with integrated Lo-Ra communication module, sensors actuators (amplifiers or interfaces) and power switches in forms of relays or semiconductor outputs [7]. Currently, the system is capable to measure three parameters: air temperature, humidity, and soil moisture, but it can be easily extended to others. Low cost and

widely available DHT11 sensor is used for air temperature and humidity, while for soil moisture measurement the special sensor solution has been developed. After collecting data from the sensors, the station sends them by LoRa communication protocol to collector, which can be about 5 km away [8]. LoRa technology uses the chirp modulation and two layers of cryptography. It can grow into a complex network of any size (LoRaWAN), transferring the data between the adjacent nodes. The LoRa module MCU has external 19dB antenna, operating at 868MHz (Europe), while other frequencies are supported worldwide. It has satisfactory power consumption. With its characteristics LoRa protocol represents a very competitive solution for purpose of PA.

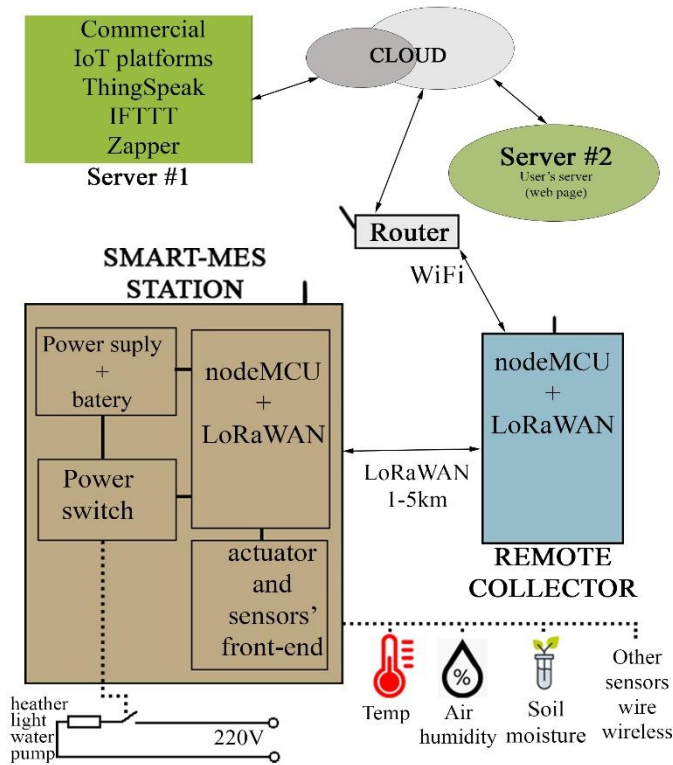


Figure 1: The general working principle of a proposed system

REMOTE COLLECTOR is a hybrid transmitter operated by the identical MCU ESP32 based node from Heltec, with integrated LoRaWAN module. Digital filters and software routines are used to ensure correct encryption of the receiving messages, under presence of different noise. In order to achieve the maximal range between two LoRaWAN nodes, the message should be as short as possible. It is because the error probability increases with the message length, and only one misinterpreted byte is enough to receive unusable content. Experiments have shown that larger communication range was achieved when sending only a few bytes compared to long descriptive strings. Additionally, in some extreme cases when it is crucial to reach the very top distance without using additional hardware, it can be done by splitting a message into one-byte sized chunks, and then sending them multiple times to gain satisfactory

probability. Both, node MCUs in the station and collector have built-in OLED display to watch on-line data or to perform adequate setting and user interfacing.

In addition of performing LoRa communication the collector has possibility to simultaneously communicated with nearby WiFi spot as it is home/office WiFi router. All communications of node MCU are duplex.

The innovative sensor device, developed during this internal experiment is a soil moisture sensor, based on capacitive principle, Figure 2. Namely, our experimentation proved that exiting low-cost soil moisture sensors, such as Capacitive soil moisture sensor v1.2 are not enough precise and, practically, unusable. The proposed soil measurement sensor is based on capacitive to PWM converter, Figure 2. It converts difference of two capacitors, $abs(C_x - C_{ref})$ into a duty cycle D of PWM signal, and thus in average value of filtered PWM signal.

$$D = T_2 / (T_1 + T_2) = f(abs(C_x - C_{ref})), VF = D * 5V + (1 - D) * 0V \quad (1)$$

The oscillator with core frequency f is driving two parallel branches RC_{ref} and RC_x , where C_x is the capacitance between the sensor electrodes, while C_{ref} is the reference capacitance. In absence of the soil C_x and C_{ref} are almost equal, $C_x = C_{ref}$, and $DELAY(RC_x) - DELAY(RC_{ref})$ is practically equal to 0, that cause minimal duty cycle and minimal average voltage. Figure 3, shows diagrams in characteristic points of proposed capacitive sensor. Amplifier amplify signal VF in a range that is readable in satisfactory resolution by microprocessor's A/D, usually in range 0-5V.

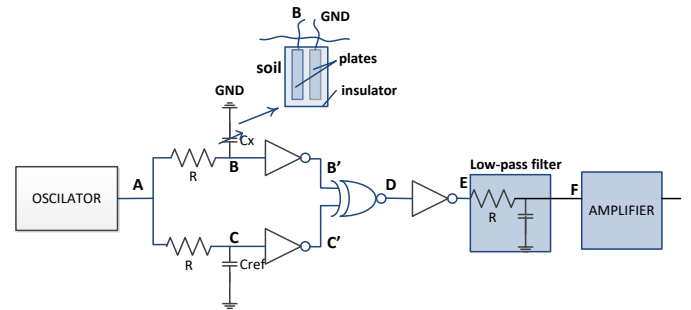


Figure 2: Principal scheme of capacitive sensor

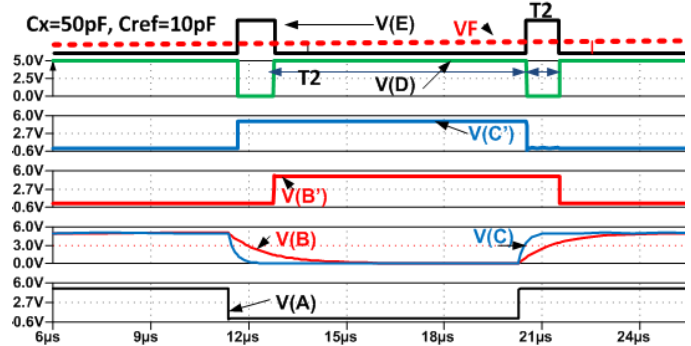


Figure 3: Diagrams in characteristic points of capacitive sensor

B. Software considerations

The software consists of firmware and cloud-based software applications. Firmware is responsible for NodeMCU general operation, sensors handling, LoRa communication, communication with ThingSpeak server and other on-board related tasks. The main component of cloud-based software is ThingSpeak application, where we setup our profile, API keys, diagram channels and other applications like TalkBack, used for handling and visualize our signals [9]. Both, nodeMCU and user can communicate with ThingSpeak. Node MCU sends data for channels (diagrams) updating and simultaneously gets the last command stored in commands queue. It is done by HTTP GET or POST protocols, implemented on a Node MCU. There are two ways to send commands to MCU via ThingSpeak. User can do it through any web browser or web application by HTTP POST and GET protocols. The same command can be set within ThingSpeak HTTP interface.

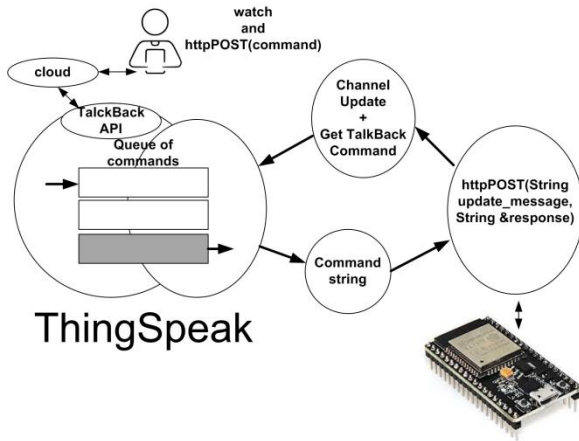


Figure 4: Software flow for sending data for diagrams and get commands

III. TESTING PROCEDURE AND OBTAINED RESULTS

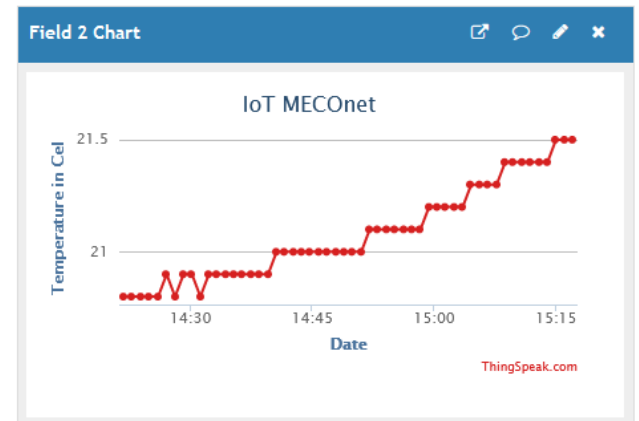
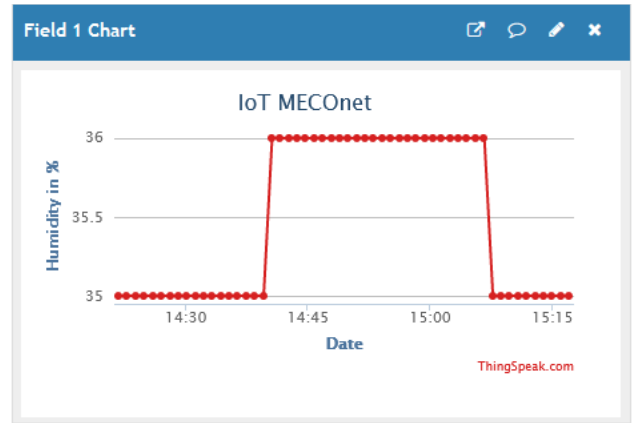
The proposed system was thoroughly tested over a period of five months and found to be precise and reliable. The length of a sending data string affected the probability of successful communication when reaching the far distance limit. So, it had to be shortened for the better results. The string was binary compressed to the length of 4 characters and reached distance of 1.5 km, which was more than enough for our testing conditions.

In term of power consumption, the following results are achieved: the device draws about 60 mA when all the circuits are powered simultaneously, adding short peaks of another 60 mA while LoRa was active.

Figures 5 show the physical look of a system and the collector serial communication output. Figure 6, shows visualization of the data sent to ThingSpeak for 3 channels, as well as visualization of active command value. Figure 7. shows interface for sending commands implemented in JavaScript code that can be very easily embedded in any web environment.



Figure 5: Station in wine yard and in micro farming environment. The collector is a small device-box located far away from station in place where Wi Fi is accessible.



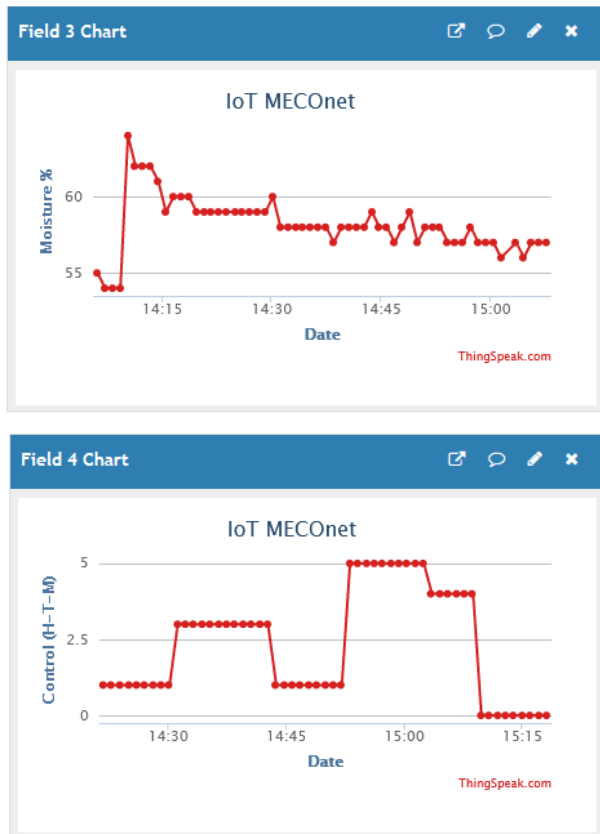


Figure 6: Charts showing: air humidity (1), air temperature (2), soil moisture (3), and control (4)

Two-way communication over the internet is of crucial significance to be able to have the fully functional IoT application, meaning that we can control our system remotely from any place in the world [6]. The Figure 7 shows an example of sending the control data from server to the MCU.

Figure 7: Sending control commands

IV. CONCLUSION

Precision agriculture is a modern approach in agro production that allows implementation of new ICT based

technologies with aim to increase production under better quality as well as to visualize, analyze and control many of the processes related to this area [2,4,5]. As example, water management is the most important [5]. In this paper, we present a system that is able to perform ICT based precision agriculture tasks. It uses low-cost components and open code platforms and as such is very suitable for developing countries, where the farmers could not purchase and install expensive commercial solutions. The main components of the proposed systems are station, collector and cloud software. The station consists of low costs sensors, node ESP32 based MCU with LoRa communication module, solar powering unit and energy devices switching unit. Collector accepts data sent by LoRa and forward them to the cloud. Its main component is identical Node ESP32 based MCU. The main cloud component is IoT server ThingSpeak, modular and suitable platform with many upgrading possibilities. The user can communicate with station in order to follow and control processes, directly via ThingSpeak or indirectly via its simple web browser application. The overall price of the system is less than 200 EUR.

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